

Calculus I Lesson 19

Quotient Rule for Derivative: $D_x \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$

$$1) \ h(x) = \frac{x^3 + 4}{x - 7}$$

$$a) \ D_x(x^3 + 4) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(x - 7) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \frac{(x - 7) \cdot D_x(x^3 + 4) - (x^3 + 4) \cdot D_x(x - 7)}{(x - 7)^2} = \underline{\hspace{2cm}}?$$

$$2) \ h(x) = \frac{x^2}{\sin x}$$

$$a) \ D_x(x^2) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$b) \ D_x(\sin x) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$c) \ h'(x) = \frac{(\sin x) \cdot D_x(x^2) - (x^2) \cdot D_x(\sin x)}{(\sin x)^2} = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$3) \ h(x) = \frac{x^2 + 5x}{2x^2 + 5x - 5}$$

$$a) \ D_x(x^2 + 5x) = \underline{\hspace{2cm}} ?$$

$$b) \ D_x(2x^2 + 5x - 5) = \underline{\hspace{2cm}} ?$$

$$c) \ h'(x) = \frac{(2x^2 + 5x - 5) \cdot D_x(x^2 + 5x) - (x^2 + 5x) \cdot D_x(2x^2 + 5x - 5)}{(2x^2 + 5x - 5)^2} = \underline{\hspace{2cm}} ?$$

$$4) \ h(x) = \frac{\cos x}{5 + \sin x}$$

$$a) \ D_x(\cos x) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(5 + \sin x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \frac{(5 + \sin x) \cdot D_x(\cos x) - (\cos x) \cdot D_x(5 + \sin x)}{(5 + \sin x)^2} = \underline{\hspace{2cm}}?$$

$$5) \ h(x) = \frac{x^2 + 1}{\sin x + \cos x}$$

$$a) \ D_x(x^2 + 1) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(\sin x + \cos x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \frac{(\sin x + \cos x) \cdot D_x(x^2 + 1) - (x^2 + 1) \cdot D_x(\sin x + \cos x)}{(\sin x + \cos x)^2} = \underline{\hspace{2cm}}?$$

$$6) \ h(x) = \frac{x^{-2}}{2x^2 + \sin x}$$

$$a) \ D_x(x^{-2}) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(2x^2 + \sin x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \frac{(2x^2 + \sin x) \cdot D_x(x^{-2}) - (x^{-2}) \cdot D_x(2x^2 + \sin x)}{(2x^2 + \sin x)^2} = \underline{\hspace{2cm}}?$$

$$7) \ h(x) = \frac{x^{2/3} + 1}{5x^2 + \sin x}$$

$$a) \ D_x(x^{2/3} + 1) = \underline{\hspace{1cm}}?$$

$$b) \ D_x(5x^2 + \sin x) = \underline{\hspace{1cm}}?$$

$$c) \ h'(x) = \frac{(5x^2 + \sin x) \cdot D_x(x^{2/3} + 1) - (x^{2/3} + 1) \cdot D_x(5x^2 + \sin x)}{(5x^2 + \sin x)^2} = \underline{\hspace{1cm}}?$$

$$8) \ h(x) = \frac{x^{-3/4} + 7x}{x^3 + \cos x}$$

$$a) \ D_x(x^{-3/4} + 7x) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$b) \ D_x(x^3 + \cos x) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$c) \ h'(x) = \frac{(x^3 + \cos x) \cdot D_x(x^{-3/4} + 7x) - (x^{-3/4} + 7x) \cdot D_x(x^3 + \cos x)}{(x^3 + \cos x)^2} = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$9) \ h(x) = \frac{x^{-5/4} + 2x^3}{x^{1/2} + \cos x}$$

$$a) \ D_x(x^{-5/4} + 2x^3) = \underline{\hspace{2cm}}?$$

$$b) \ D_x(x^{1/2} + \cos x) = \underline{\hspace{2cm}}?$$

$$c) \ h'(x) = \frac{(x^{1/2} + \cos x) \cdot D_x(x^{-5/4} + 2x^3) - (x^{-5/4} + 2x^3) \cdot D_x(x^{1/2} + \cos x)}{(x^{1/2} + \cos x)^2} = \underline{\hspace{2cm}}?$$

